EXPERIMENT 1 – AJAY KARTHICK

AIM

To calculate the cutoff frequencies, propagation constant and attenuation constant of a rectangular waveguide, and to plot the TE10 and TM11 field patterns using MATLAB.

SOFTWARE REQUIRED

MATLAB

PROCEDURE

1. Open MATLAB and create a new script.
2. Enter the waveguide dimensions, operating frequency and speed of light.
3. Calculate the cutoff frequencies, propagation constant and attenuation constant.
4. Plot the electric and magnetic field patterns for the specified modes.
5. Plot the TE10 propagation constant against frequency and observe the output.

CODE

```
clc; clear; close all;

c = 3e8;
f = 10e9;
a = 22.86e-3;
b = 10.16e-3;

fc = c/(2*a);
ft = (c/2)*sqrt((1/a)^2 + (1/b)^2);
k = 2*pi*f/c;
beta = sqrt(k^2 - (2*pi*fc/c)^2);
al = sqrt((2*pi*ft/c)^2 - k^2);

fprintf('TE10 cutoff frequency = %.3f GHz\\n', fc/1e9);
fprintf('TM11 cutoff frequency = %.3f GHz\\n', ft/1e9);
fprintf('Propagation constant beta = %.3f rad/m\\n', beta);
fprintf('Attenuation constant alpha = %.3f Np/m\\n', al);

x = linspace(0,a,100);
y = linspace(0,b,100);
[X,Y] = meshgrid(x,y);

figure('Color','w');
subplot(2,2,1);
imagesc(x*1e3,y*1e3,sin(pi*X/a));
axis xy; colorbar; title('TE10 E-Field');
xlabel('x (mm)'); ylabel('y (mm)');

subplot(2,2,2);
imagesc(x*1e3,y*1e3,cos(pi*X/a));
axis xy; colorbar; title('TE10 H-Field');
xlabel('x (mm)'); ylabel('y (mm)');

subplot(2,2,3);
imagesc(x*1e3,y*1e3,sin(pi*X/a).*sin(pi*Y/b));
```

```
axis xy; colorbar; title('TM11 E-Field');
xlabel('x (mm)'); ylabel('y (mm)');

subplot(2,2,4);
imagesc(x*1e3,y*1e3,sin(pi*X/a));
axis xy; colorbar; title('TM11 H-Field Pattern');
xlabel('x (mm)'); ylabel('y (mm)');
colormap(jet);

F = linspace(1e9,20e9,80);
B = nan(size(F));
q = F > fc;
B(q) = 2*pi*F(q)/c .* sqrt(1 - (fc./F(q)).^2);

figure('Color','w');
plot(F/1e9,B,'b','LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
ylabel('Beta (rad/m)');
title('TE10 Propagation Constant');
```

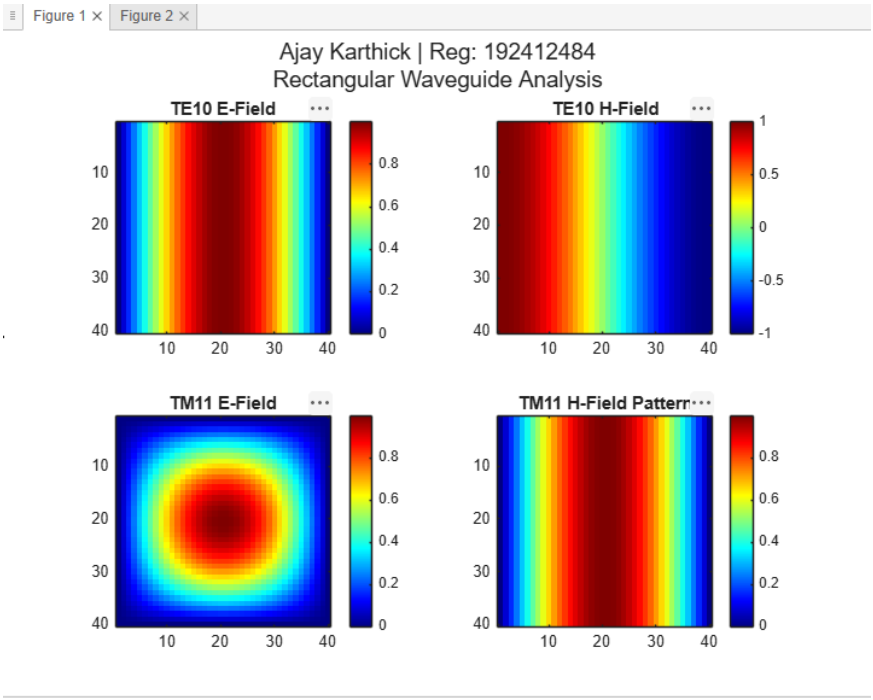
OUTPUT

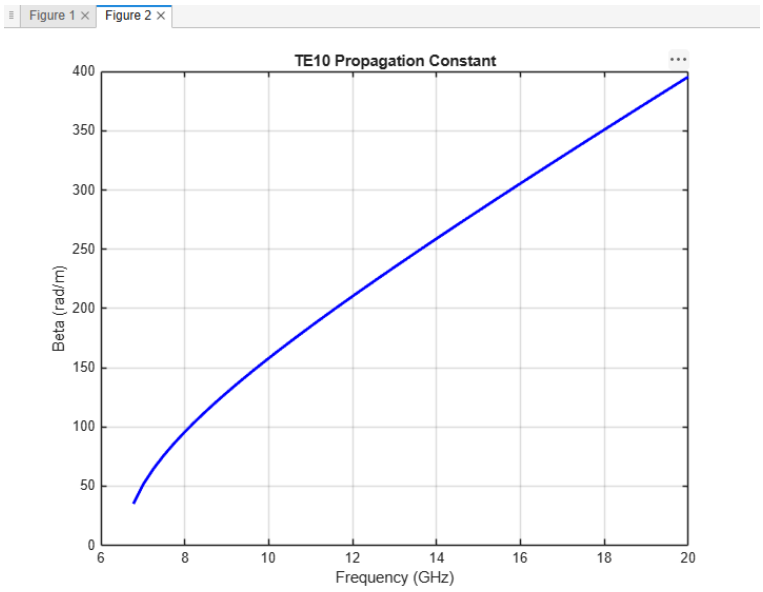
Editor

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/MATLAB Drive/untitled.m

```
1   clc; clear; close all;
2   c=3e8; f=10e9;
3   a=22.86e-3; b = 10.16e-3;
4   fc=c/(2*a);
5   ft=c/2*sqrt((1/a)^2+(1/b)^2);
6   k=2*pi*f/c;
7   beta=sqrt(k^2-(2*pi*fc/c)^2);
8   al=sqrt((2*pi*ft/c)^2-k^2);
9   fprintf('Ajay Karthick | Reg: 192412484\n');
10  fprintf('TE10 fc=%.3f GHz\n',fc/1e9);
11  fprintf('Beta=%.2f rad/m\n',beta);
12  x=linspace(0,a,40); y=linspace(0,b,40);
13  [X,Y]=meshgrid(x,y);
14  figure('Color','w');
15  sgtitle({'Ajay Karthick | Reg: 192412484', ...
16  'Rectangular Waveguide Analysis'});
17  subplot(2,2,1); imagesc(sin(pi*X/a)); colorbar; title('TE10 E-Field');
18  subplot(2,2,2); imagesc(cos(pi*X/a)); colorbar; title('TE10 H-Field');
19  subplot(2,2,3); imagesc(sin(pi*X/a).*sin(pi*Y/b)); colorbar; title('TM11 E-Field');
20  subplot(2,2,4); imagesc(sin(pi*X/a)); colorbar; title('TM11 H-Field Pattern');
21  colormap(jet);
22  F=linspace(1e9,20e9,80); B=nan(size(F)); q=F>fc;
23  B(q)=2*pi*F(q)/c.*sqrt(1-(fc./F(q)).^2);
24  figure('Color','w'); plot(F/1e9, B, 'b', 'LineWidth',2); grid on;
25  xlabel('Frequency (GHz)'); ylabel('Beta (rad/m)');
26  title('TE10 Propagation Constant');
```





REFLECTION

This experiment helped me understand rectangular waveguide cutoff frequencies, propagation and attenuation constants, and the visualization of electromagnetic field patterns using MATLAB.

RESULT

The cutoff frequencies, propagation constant and attenuation constant were calculated, and the field patterns and TE10 propagation-constant graph were plotted using MATLAB.

Date
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Experiment - 1

$a = 22.86\,mm = 0.02286\,m$
$b = 10.16\,mm = 0.01016\,m$
$f = 10\,GHz$
$C = 3 \times 10^8\,m/s$

$$f_c = C/2a \Rightarrow 3\times10^8 / 2\times0.02286$$

$$\boxed{f_c = 6.562\,GHz}$$

$$f_t = (C/2) \times \sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2}$$

$$\Rightarrow \frac{3\times10^8}{2} \times \sqrt{\frac{1}{(0.02286)^2} + \frac{1}{(0.01016)^2}}$$

$$\boxed{f_t = 16.156\,GHz}$$

$$K = 2\pi f/C$$

$$K = (2 \times 3.1416 \times 10\times10^9)/(3\times10^8)$$

$$\boxed{K = 209.44\,rad/m}$$

$$\beta = \sqrt{k^2 - \left(\frac{2\pi f_c}{C}\right)^2}$$

$$\boxed{\beta = 158.05\,rad/m}$$

$$\alpha = \sqrt{(2\times3.146\times16.156\times10^8/3\times10^8)^2 - (209.44)^2}$$

$$\boxed{\alpha = 265.77\,Np/m}$$